Exploring shapely

In […
```
import shapely

p = shapely.Point(1,2)
```

In […
```
p
```

Out[…

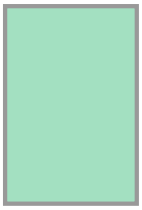
That isn't a placeholder, the point is being displayed on the screen.

In […
```
poly = shapely.Polygon([(0,0),(2,0),(2,3),(0,3),(0,
poly.contains(p)
```

Out[… True

In […
```
poly
```

Out[…



There are various points that can stand in for the whole object: `centroid` and `representative_point`.

In […
```
poly.centroid
```

Out[…

In […
```
print(poly.centroid)
```

```
POINT (1 1.5)
```

In […
```
type(poly.centroid)
```

Out[…
```
shapely.geometry.point.Point
```

In […
```
rp = poly.representative_point()
print(rp)
```

```
POINT (1 1.5)
```

In […
```
type(rp)
```

Out[…
```
shapely.geometry.point.Point
```

In […
```
q = shapely.Point(4,5)
q.distance(p)
```

Out[…
```
4.242640687119285
```

A point has an attribute `coords` whose type is:

In […
```
type(p.coords)
```

Out[…
```
shapely.coords.CoordinateSequence
```

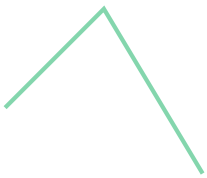
An easy way to get at a `CoordinateSequence` is to turn it into a `list`:

In […
```
list(p.coords)
```

Out[…
```
[(1.0, 2.0)]
```

In […
```
r = shapely.Point(7,0)
LS = shapely.LineString([p,q,r])
LS
```

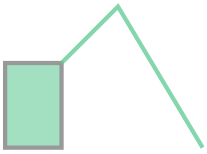
Out[…



In […

```
LS.union(poly)
```

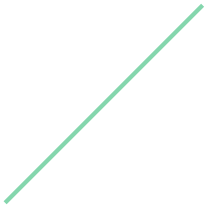
Out[…



In […

```
LS.intersection(poly)
```

Out[…



In […

```
type(LS)
```

Out[… shapely.geometry.linestring.LineString

In […

```
type(LS.coords)
```

Out[… shapely.coords.CoordinateSequence

In […

```
list(LS.coords)
```

Out[… [(1.0, 2.0), (4.0, 5.0), (7.0, 0.0)]

There is a much more complicated way to dig down into a `CoordinateSequence`:

In […

```
LS.coords[:][0]
```

Out[… (1.0, 2.0)

The brackets colon notation in Python means to make a copy of a list. Here it converts a `CoordinateSequence` object into a

`list`.

In […
```
type(LS.coords[:])
```

Out[…
```
list
```

Of course, a list can have more than one element. Hence the `[0]`.

In […
```
print(poly)
```

```
POLYGON ((0 0, 2 0, 2 3, 0 3, 0 0))
```

Notice the double parentheses. The reason for this is that it is allowed for a polygon to have holes. (For example, a state might contain lakes).

In […
```
border = [(-180, 90), (-180, -90), (180, -90), (180
holes = [[(-170, 80), (-170, -80), (170, -80), (170
frame = shapely.Polygon(shell=border, holes=holes)
print(frame)
```

```
POLYGON ((-180 90, -180 -90, 180 -90, 180 90, -180
90), (-170 80, -170 -80, 170 -80, 170 80, -170 80))
```

In […
```
frame
```

Out[…

In […
```
print(f"Polygon centroid: {frame.centroid}")
print(f"Polygon Area: {frame.area}")
print(f"Polygon Bounding Box: {frame.bounds}")
```

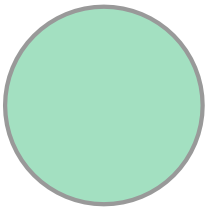
```
print(f"Polygon Exterior: {frame.exterior}")
print(f"Polygon Exterior Length: {frame.exterior.le
```

```
Polygon centroid: POINT (0 0)
Polygon Area: 10400.0
Polygon Bounding Box: (-180.0, -90.0, 180.0, 90.0)
Polygon Exterior: LINEARRING (-180 90, -180 -90, 18
0 -90, 180 90, -180 90)
Polygon Exterior Length: 1080.0
```

In […

```
# Circle (using a buffer around a point)
point = shapely.Point((0,0))
point.buffer(1)
```

Out[…



In […